

1. 2014Q19

Which of the following is **not** an application of DNA fingerprinting?

- A. Forensic science
- B. Screening for genetic diseases
- C. Sequencing of the human genome
- D. Identification of Chinese medicines

2. 2020Q17

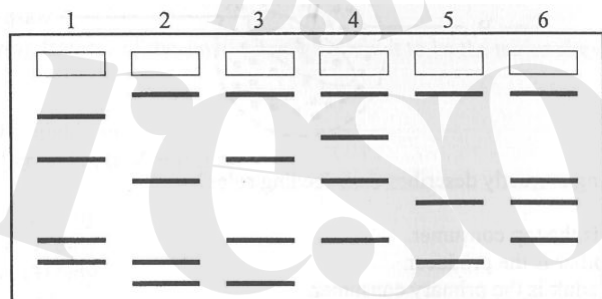
‘Transgenic organisms produced by recombinant DNA technology have more potential in terms of evolution than those produced by traditional breeding.’

Which of the following is the best reason for this?

- A. Recombinant DNA technology creates new species.
- B. Recombinant DNA technology produces new phenotypes.
- C. Recombinant DNA technology transfers genes within a species.
- D. Recombinant DNA technology transfers genes between different species.

3. 2023Q26

The DNA fingerprints below show the screening of a rare genetic disease in a family of six members:



The onset of the disease occurs in adulthood. Members 2 and 6 have already shown the symptoms of disease.

Which member is likely to have inherited this disease too?

- A. 1
- B. 3
- C. 4
- D. 5

LQ

1. 2022Q9biii

- b) Bacterium R is a pathogenic bacterium which possesses an antibiotic resistance gene. Its gene product can break down antibiotic X. Scientists have suggested a new approach to fight against bacterium R. This new approach involves the use of a synthetic polynucleotide which binds to the mRNA transcribed from the antibiotic resistance gene. The expression of the gene is then inhibited. By administering this synthetic polynucleotide together with antibiotic X, bacterium R can be killed.
- iii) Suggest **one** advantage of using synthetic polynucleotide to fight against antibiotic resistant bacteria. (1 mark)

Ans

2014Q19: C (18%)

2020Q17: D (58%)

2023Q26: C (81%)

2022Q9biii

9 (b) (iii) any **one** of the followings:

- can design a new synthetic polynucleotide any time once we know the sequence of antibiotic resistance gene (1)
- the action is very specific (1) as it targets the resistance gene

(1)